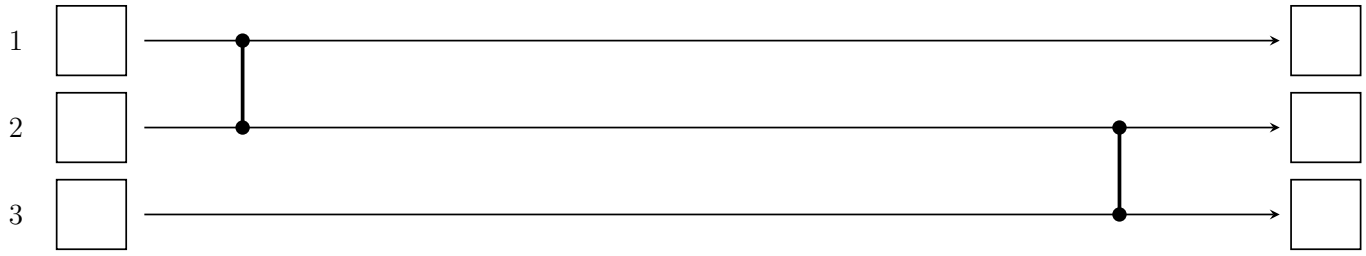
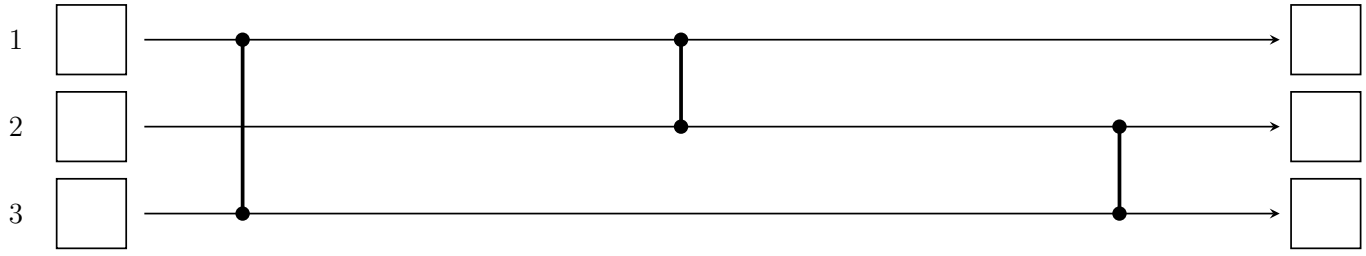
Move from left to right. At each bar, compare only the two lanes with black dots: the smaller value goes to the upper lane and the larger to the lower lane. Ties may stay put. Keep every bar fixed for every start; move cards only at bars. A machine sorts when the final values never get smaller from lane 1 down.

Problem 1: For each machine, find every starting order of 1, 2, 3 that it sorts. Which machines sort all starting orders?





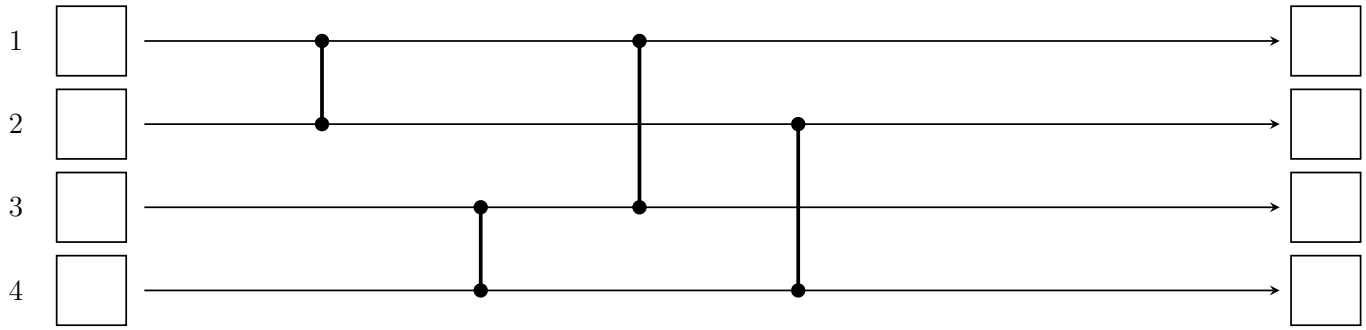


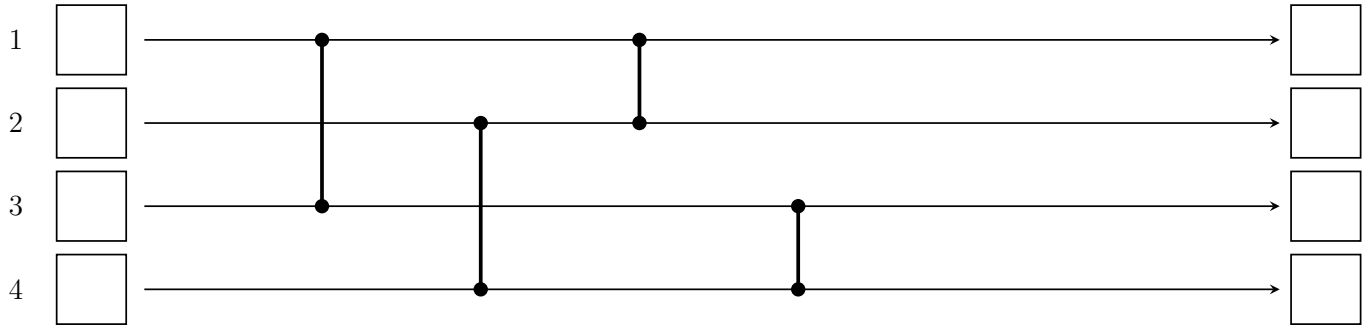
Problem 2: Build a three-lane machine that sorts every starting order of 1, 2, 3. Use as few bars as you can, and explain why fewer bars cannot work.

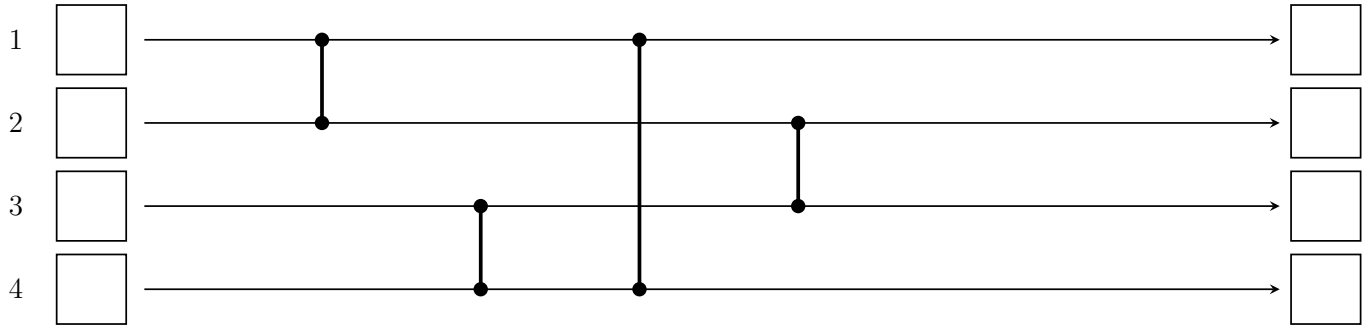




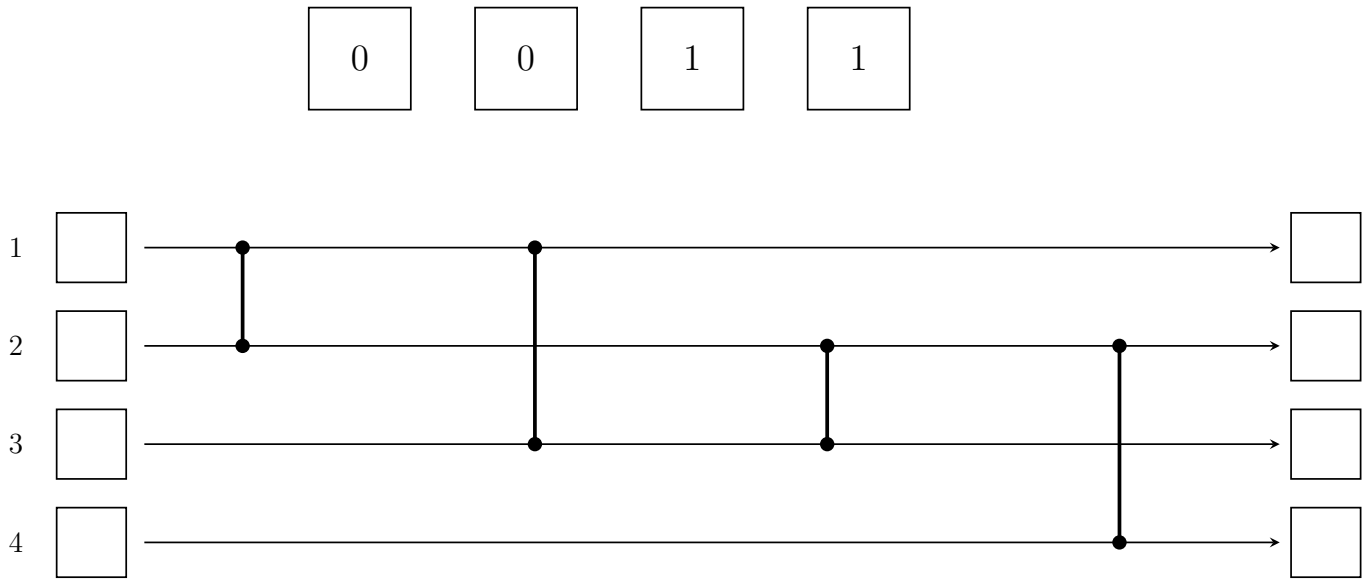
Problem 3: For each machine, add one bar at the right end so that it sorts every starting order of 1, 2, 3, 4. If one new bar cannot do it, explain why.







Problem 4: Use the cards 0, 0, 1, 1. Find every different starting order and test it on this machine. Decide whether the machine also sorts every other four-card start made of 0s and 1s.



Problem 5: Build a four-lane machine that sorts every start made of 0s and 1s. Make a complete record of the starting arrangements you checked.



Problem 6: Build a four-lane machine that sorts every order of 1, 2, 3, 4. Every bar must join neighboring lanes: 1 and 2, 2 and 3, or 3 and 4. Use as few bars as you can, and explain why fewer cannot work.

